

Problem

Euclid's Identity states that for any two positive integers A and B, there exist integers x and y such that:  $Ax + By = D$

where D is the Greatest Common Divisor (GCD) of A and B.

Given two positive integers A and B, find:  $x \ y \ D = \text{gcd}(A, B)$

such that:

$Ax + By = D$

If multiple pairs (x, y) satisfy the equation, choose the pair with

the minimum value of:  $|x| + |y|$

If there is still more than one pair, choose the pair for which:

$x \leq y$

Input Format

A B

Constraints

$1 \leq A, B \leq 10^9$

Output Format

x y D D is gcd(A, B) Ax + By = D (x, y) satisfies the minimum-selection rules.

Submissions

Leaderboard

Discussions

Sample Input 0

4 6

Sample Output 0

-1 1 2

Change Theme

Language

Python 3



```
1 # Enter your code here. Read input from STDIN.
2 A, B = map(int, input().split())
3 def egcd(a, b):
4     if b == 0:
5         return a, 1, 0
6     d, x1, y1 = egcd(b, a % b)
7     x = y1
8     y = x1 - (a // b) * y1
9     return d, x, y
10 D, x, y = egcd(A, B)
11 bx = x
12 by = y
13 k = round(-x * D / B)
14 for i in range(k - 2, k + 3):
15     nx = x + i * (B // D)
16     ny = y - i * (A // D)
17     if abs(nx) + abs(ny) < abs(bx) + abs(by):
18         bx = nx
19         by = ny
20     elif abs(nx) + abs(ny) == abs(bx) + abs(by):
21         if nx <= ny:
22             bx = nx
23             by = ny
24 print(bx, by, D)
```

Line: 23 Col: 20



Upload Code as File

Run Code

Submit Code



Test against custom input

# Congratulations

You solved this challenge. Would you like to challenge your friends?



Test case 0

Compiler Message

Success